MANAGEMENT INFORMATION SYSTEMS
ICT2162
LECTURE 07

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Managing Knowledge and Artificial Intelligence

What We Discuss Today?

- Role of knowledge management(KM)
- Knowledge management programs in business
- Types of systems used for enterprise-wide knowledge management
- How KM systems provide value for businesses
- Major types of knowledge work systems
- Business benefits of using intelligent techniques for knowledge management

Designing Drugs Virtually

- Problem:
  - Ineffective and complicated drug discovery process
- Solutions:
  - Use AI-driven structure-based design to identify potential molecules for new drugs.
  - Use of **digital twins in healthcare** to simulate effects.
- **Illustrates IT's role** in accelerating drug discovery, improving efficiency, and reducing costs.

The Role of Knowledge Management Systems in Business

- Knowledge management and collaboration platforms are among the **fastest-growing investment areas in both corporate and government sectors**.

- **Knowledge economy (2025):**
- Over **30% of global GDP** now derives from the information and digital knowledge sectors.
- Intangible assets (knowledge, brand, reputation, proprietary processes) remain **core drivers of enterprise value**.

- **Well-executed AI-driven KM projects** can yield extraordinary ROI.

What is Knowledge: How It is Getting Generated

- Data
- Data → Information
- Information → Knowledge
- Knowledge → Wisdom

- Data → information → knowledge → Wisdom

Important Dimensions of Knowledge

- Knowledge is a firm asset
  - Intangible
  - Creation of knowledge from data, information, requires organizational resources
  - As it is shared, experiences network effects because its value increases as more people share it. Knowledge today includes **AI-generated knowledge and knowledge graphs**

- Knowledge has different forms
  - Tacit (human expertise, experience)
  - Explicit (codified)
  - AI-generated (machine-assisted insights)

Important Dimensions of Knowledge

- Knowledge has a location.
  - Cognitive event
  - Both social and individual
  - “Sticky” (hard to move), situated (enmeshed in firm’s culture), contextual (works only in certain situations)
- Knowledge is situational.
  - Conditional :Knowing when to apply procedure
  - Contextual : Knowing circumstances to use certain tool

The Role of Knowledge Management Systems in Business

- Modern firm's value depends on the firm's ability to create and manage knowledge
- Knowing how to do things
  - effectively and efficiently
  - in ways others cannot duplicate is prime source of profit and competitive advantage
    - For example, Having a unique build-to-order production system

Organizational Learning

- Process in which organizations learn
  - ✓ Gain experience through
    - collection of data
    - careful measurement of planned activities
    - trial and error (experimentation)
    - feedback from customers and the environment
  - ✓ Adjust behavior to reflect experience
    - Create new business processes with AI- driven insights.
    - Change patterns of management decision making.

Knowledge Management Value Chain

- Knowledge management

  - Set of business processes developed in an organization to
    - Create
    - Store
    - Transfer and
    - apply knowledge into its business processes

Knowledge Management Value Chain

- Knowledge management value chain:
  - Each stage adds value to raw data and information as they are transformed into usable knowledge.

  ✓ **Knowledge Acquisition**
  ✓ **Knowledge Storage**
  ✓ **Knowledge Dissemination**
  ✓ **Knowledge Application**

Knowledge Management Value Chain

Knowledge Acquisition

- Documenting tacit and explicit knowledge
  - Storing documents, reports, presentations, best practices
  - Unstructured documents (e.g., e- mails)
  - Developing online expert networks
- Creating knowledge
- Tracking data from TPS and external sources

Knowledge Management Value Chain

Knowledge Storage

- Databases
- Document management systems
- Role of management:
  - Support development of planned knowledge storage systems
  - Encourage development of corporate wide schemas for indexing documents
  - Reward employees for taking time to update and store documents properly

Knowledge ManagementValue Chain

Knowledge Dissemination

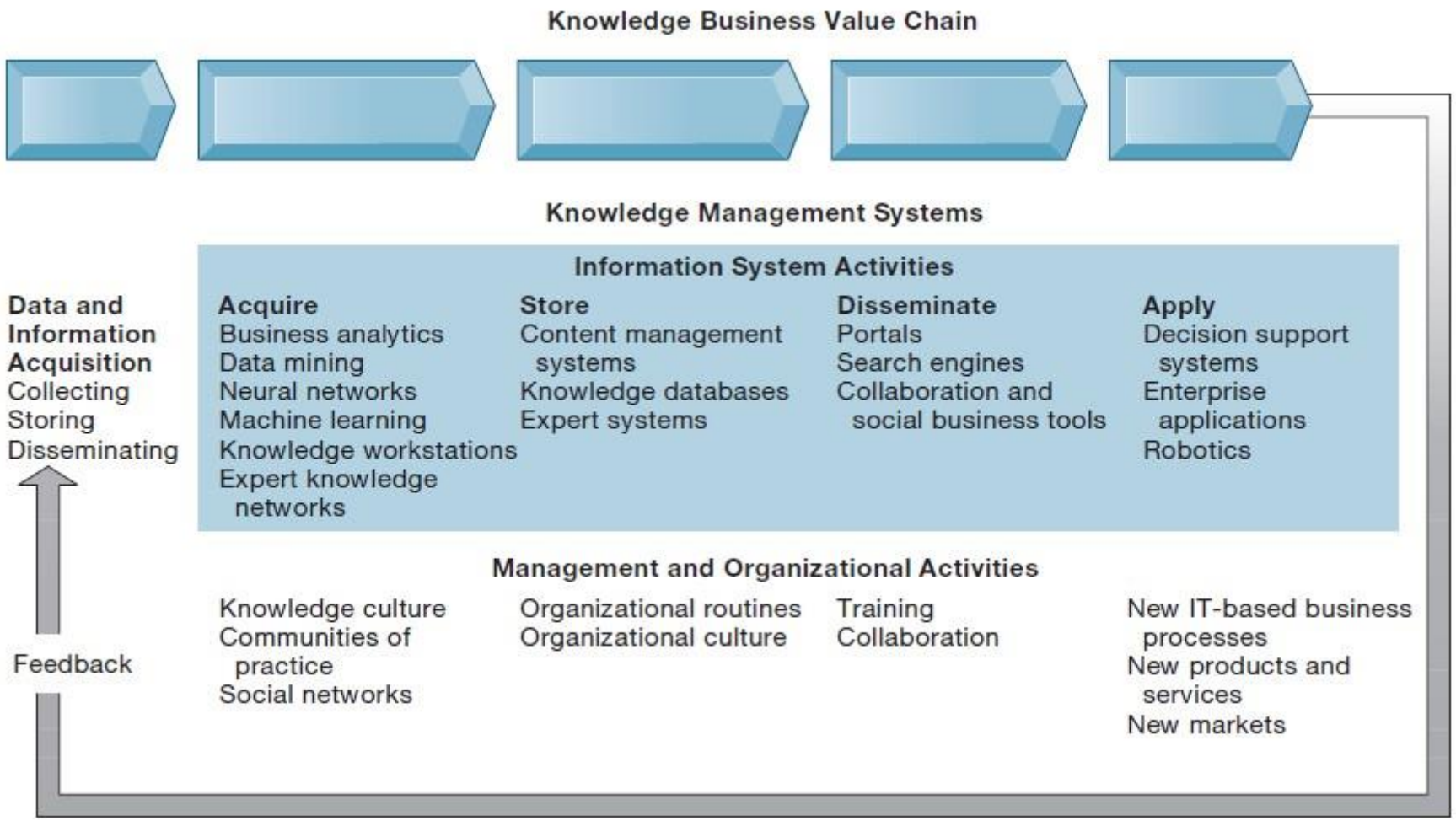
- Portals, wikis
- E-mail, instant messaging
- Search engines
- Collaboration tools
- Training programs, informal networks, and shared management experience help managers focus attention on important information.

Knowledge ManagementValue Chain

Knowledge Application

- To provide return on investment, organizational knowledge must become systematic part of management decision making and become situated in decision- support systems
  - New business practices
  - New products and services
  - New markets

Knowledge Management Value Chain



Organizational Roles and Responsibilities

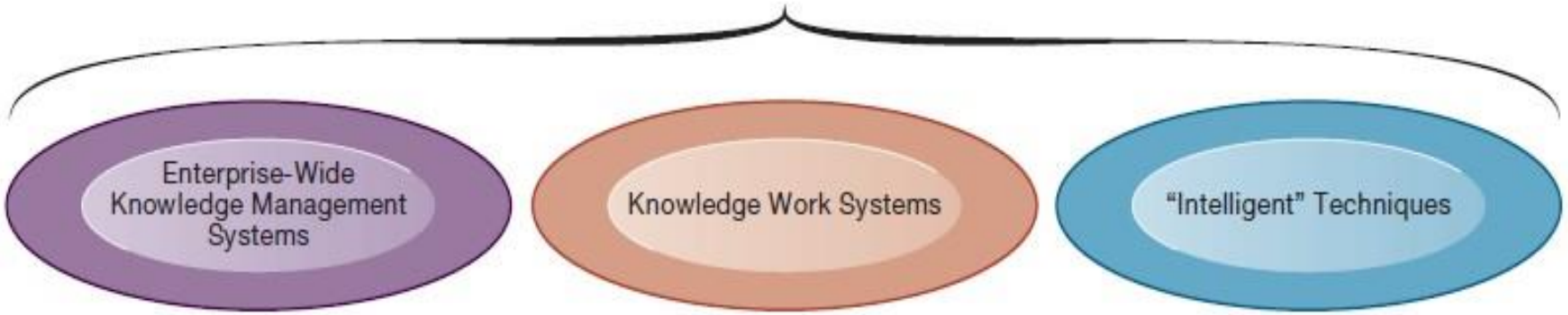
- Chief knowledge officer executives
- Dedicated staff / knowledge managers
- Communities of Practice (COPs)
  - Informal social networks of professionals and employees within and outside firm who have similar work-related activities and interests
  - Activities include education, online newsletters, sharing experiences and techniques
  - Facilitate reuse of knowledge, discussion
  - Reduce learning curves of new employees

Three Major Types of Knowledge Management Systems

1. **Enterprise-wide knowledge management systems**
   - General-purpose firm-wide efforts to collect, store, distribute, and apply digital content and knowledge
2. **Knowledge work systems (KWS)**
   - Specialized systems built for engineers, scientists, other knowledge workers charged with discovering and creating new knowledge
3. **Intelligent techniques**
   - Diverse group of techniques such as data mining used for various goals: discovering knowledge, distilling knowledge, discovering optimal solutions

Three Major Types of Knowledge Management Systems

There are three major categories of knowledge management systems, and each can be broken down further into more specialized types of knowledge management systems.



General-purpose, integrated, firmwide efforts to collect, store, disseminate, and use digital content and knowledge

Enterprise content management systems
Collaboration and social tools
Learning management systems

Specialized workstations and systems that enable scientists, engineers, and other knowledge workers to create and discover new knowledge

Computer-aided design (CAD)
Virtual reality

Tools for discovering patterns and applying knowledge to discrete decisions and knowledge domains

Data mining
Neural networks
Expert systems
Machine learning
Natural language processing
Computer vision systems
Robotics
Genetic algorithms
Intelligent agents

Enterprise-wide Knowledge Management Systems

Three Major Types of Knowledge in Enterprise

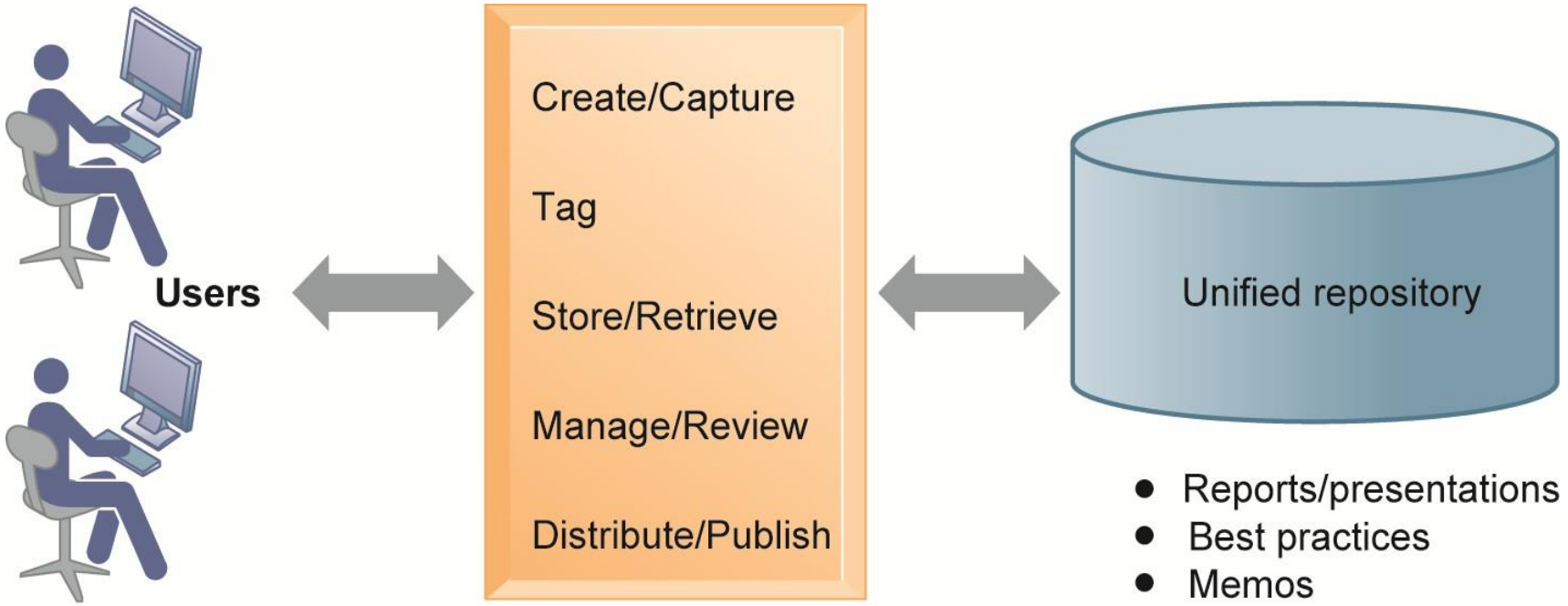
1. **Structured Documents**
   - Reports, Presentations
   - Formal Rules
2. **Semi structured Documents**
   - E-mails, Videos
3. **Unstructured, Tacit Knowledge**

- 80% Of An Organization's Business Content Is Semi structured Or Unstructured

Enterprise Content Management Systems

- Help capture, store, retrieve, distribute, preserve
  - Documents, reports, best practices
  - Semi structured knowledge (e-mails)
- Bring in external sources
  - News feeds, research
- Tools for communication and collaboration
  - Blogs, wikis, and so on

An Enterprise Content Management System



- Reports/presentations
- Best practices
- Memos
- Digital slides
- E-mail
- Graphics
- Video
- News feeds
- Wikis, blogs
- Employee profiles

Enterprise Content Management Systems

- Key problem—Developing taxonomy
  - Knowledge objects must be tagged with categories for retrieval
- Digital asset management systems
  - Specialized content management systems for classifying, storing, managing unstructured digital data
  - Photographs, graphics, video, audio

Locating and Sharing Expertise

- Provide online directory of corporate experts in well-defined knowledge domains
- Search tools enable employees to find appropriate expert in a company
- Social networking and social business tools for finding knowledge outside the firm
- Saving, tagging, sharing Web pages

Learning Management Systems (LMS)

- Provide tools for management, delivery, tracking, and assessment of various types of employee learning and training
- Support multiple modes of learning
  - CD-ROM, Web-based classes, online forums, live instruction, and so on
- Automates selection and administration of courses
- Assembles and delivers learning content
- Measures learning effectiveness
- AI tutors adaptive learning systems, and personalized pathways.
- Massively open online courses (MOOCs)
  - Web course open to large numbers of participants

Knowledge Work Systems (KWS)

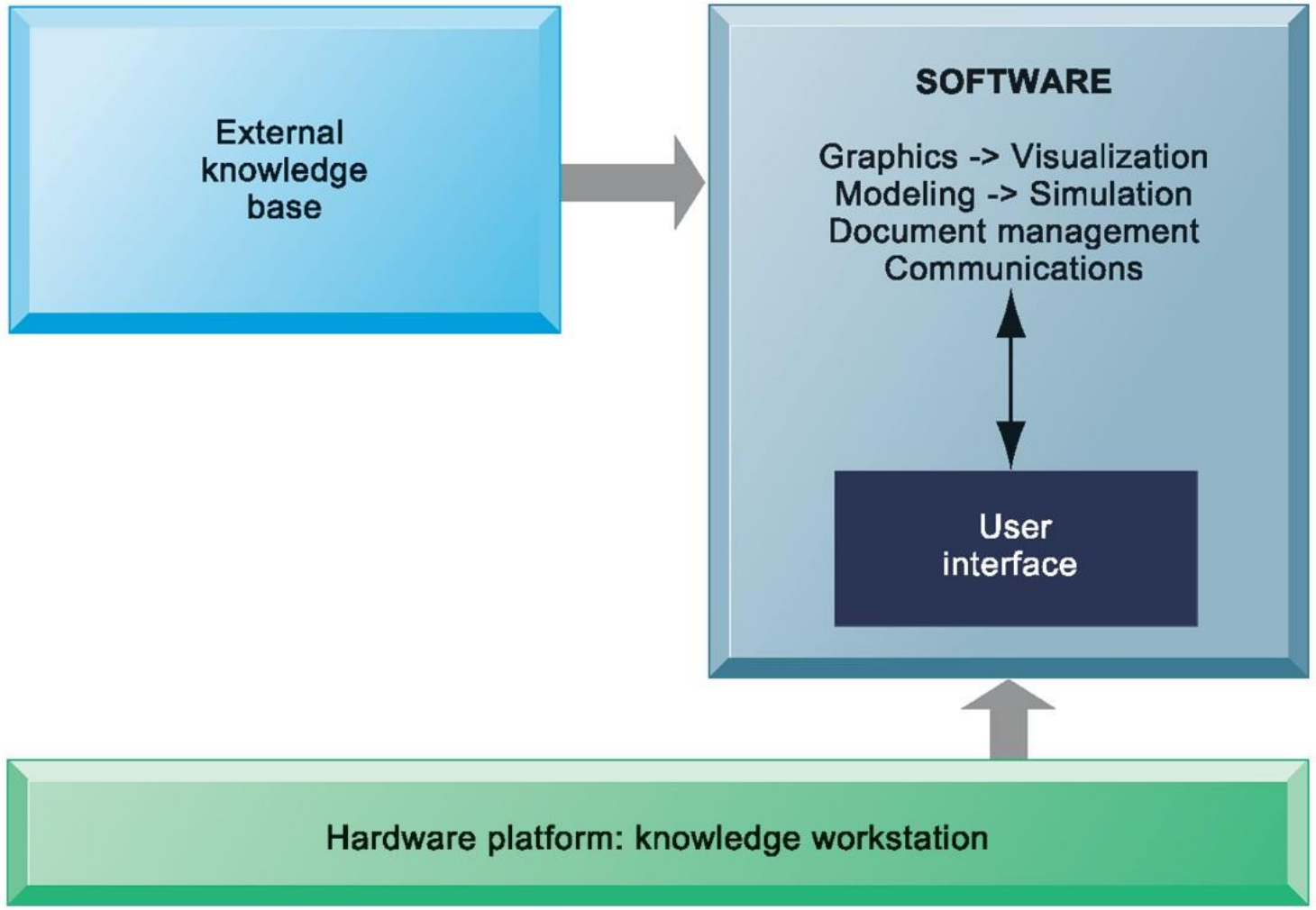
Knowledge Work Systems

- Knowledge work systems
  - Systems for knowledge workers to help create new knowledge and integrate that knowledge into business
- Knowledge workers
  - Researchers, designers, architects, scientists, engineers who create knowledge for the organization
  - Three key roles:
    - Keeping organization current in knowledge
    - Serving as internal consultants regarding their areas of expertise
    - Acting as change agents, evaluating, initiating, and promoting change projects

Requirements of Knowledge Work Systems

- Sufficient computing power for graphics, complex calculations
- Powerful graphics and analytical tools
- Communications and document management
- Access to external databases
- User-friendly interfaces
- Optimized for tasks to be performed (design engineering, financial analysis)

Requirements of Knowledge Work Systems



Examples of Knowledge Work Systems

- CAD (Computer-Aided Design):
  - Creation of engineering or architectural designs
  - 3-D printing(Additive manufacturing)
- Virtual reality systems:
  - Simulate real-life environments
  - 3-D medical modeling for surgeons
  - Augmented reality (AR) systems
  - VRML
- Investment workstations:
  - Streamline investment process and consolidate
  - internal, external data for brokers, traders, portfolio managers

Intelligent Techniques

Intelligent Techniques

- **Intelligent techniques(Traditional Approach):** Used to capture individual and collective knowledge and to extend knowledge base
  - To capture tacit knowledge: Expert systems, case- based reasoning, fuzzy logic
  - Knowledge discovery: Neural networks and data mining
  - Generating solutions to complex problems: Genetic algorithms
  - Automating tasks: Intelligent agents

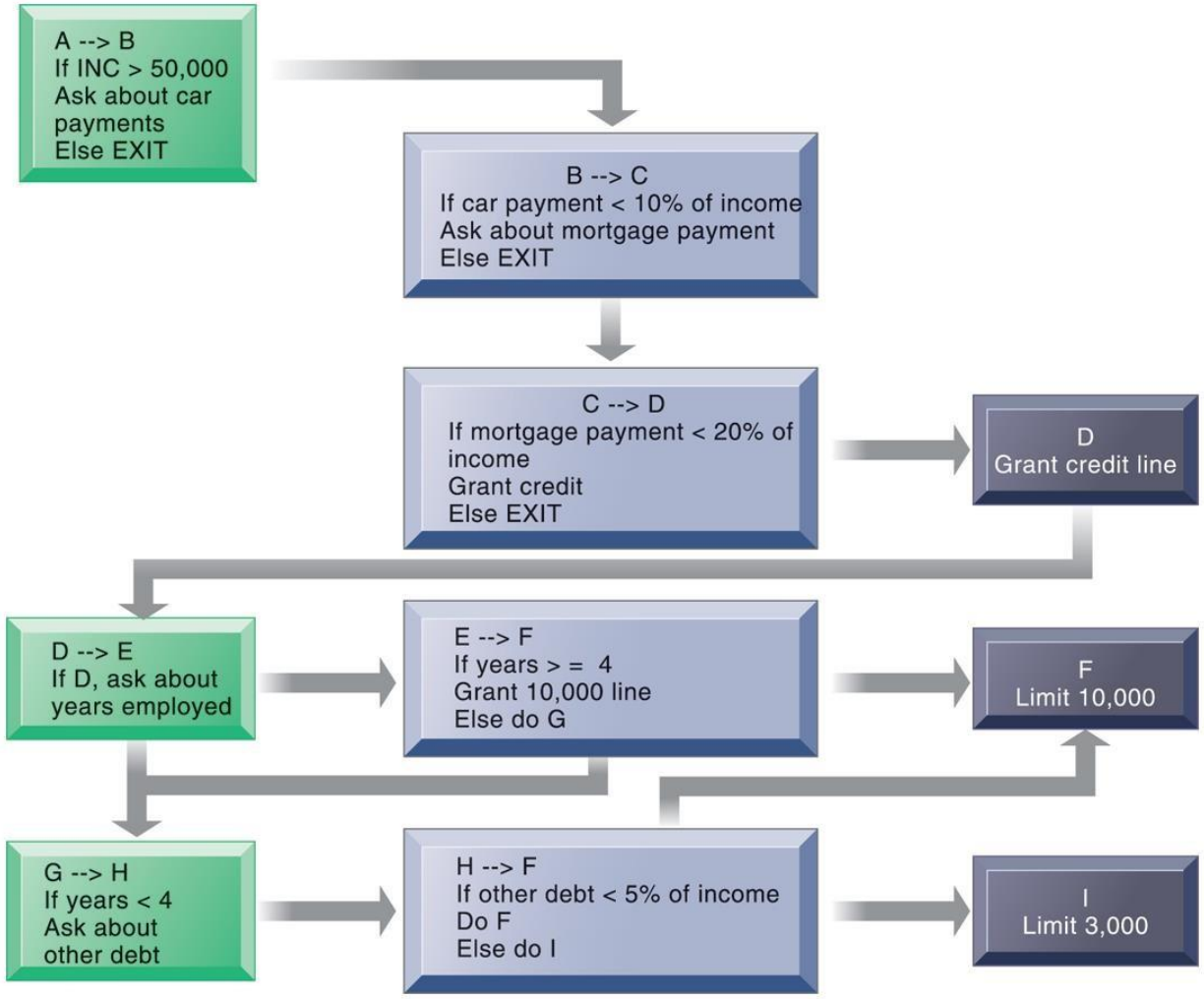
Intelligent Techniques

- **Intelligent techniques(Modern Approach):**
  - Machine learning and deep learning (transformers, reinforcement learning)
  - Generative AI (GPT-5, Llama, Gemini, Claude)
  - Multimodal AI (text, image, video, speech)
  - Retrieval-Augmented Generation (RAG)
  - Autonomous AI agents (AutoGPT, BabyAGI, crew AI)

Expert Systems

- Capture tacit knowledge in very specific and limited domain of human expertise
- Capture knowledge of skilled employees as set of rules in software system that can be used by others in organization
- Typically perform limited tasks that may take a few minutes or hours, for example:
  - Diagnosing malfunctioning machine
  - Determining whether to grant credit for loan
- Used for discrete, highly structured decision making

Rules in an Expert System

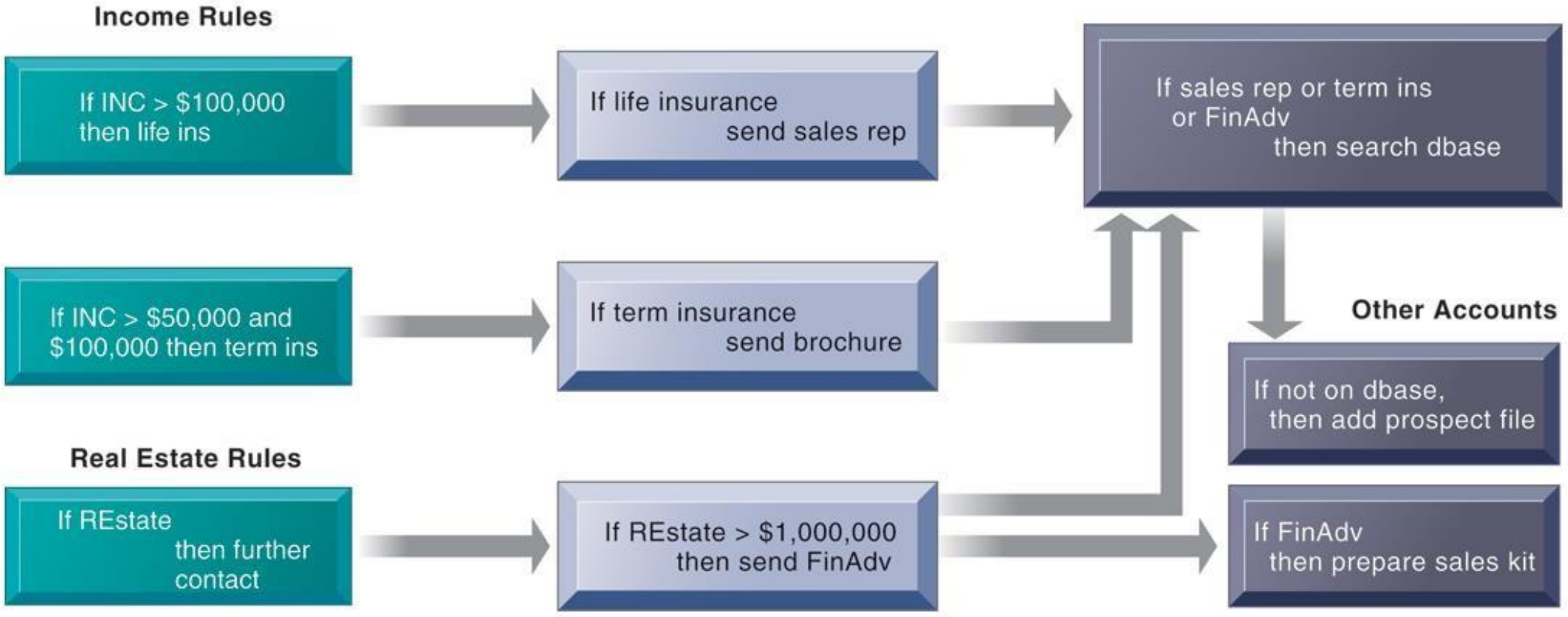


An expert system contains a number of rules to be followed. The rules are interconnected; the number of outcomes is known in advance and is limited; there are multiple paths to the same outcome; and the system can consider multiple rules at a single time.The rules illustrated are for simple credit-granting expert systems.

How Expert Systems Work

- Knowledge base:
  - Set of hundreds or thousands of rules
- Inference engine:
  - Strategy used to search knowledge base
    - **Forward chaining**: Inference engine begins with information entered by user and searches knowledge base to arrive at conclusion
    - **Backward chaining**: Begins with hypothesis and asks user questions until hypothesis is confirmed or disproved

Inference Engines in Expert Systems



An inference engine works by searching through the rules and "firing" those rules that are triggered by facts gathered and entered by the user. Basically, a collection of rules is similar to a series of nested IF statements in a traditional software program; however, the magnitude of the statements and degree of nesting are much greater in an expert system.

Expert Systems

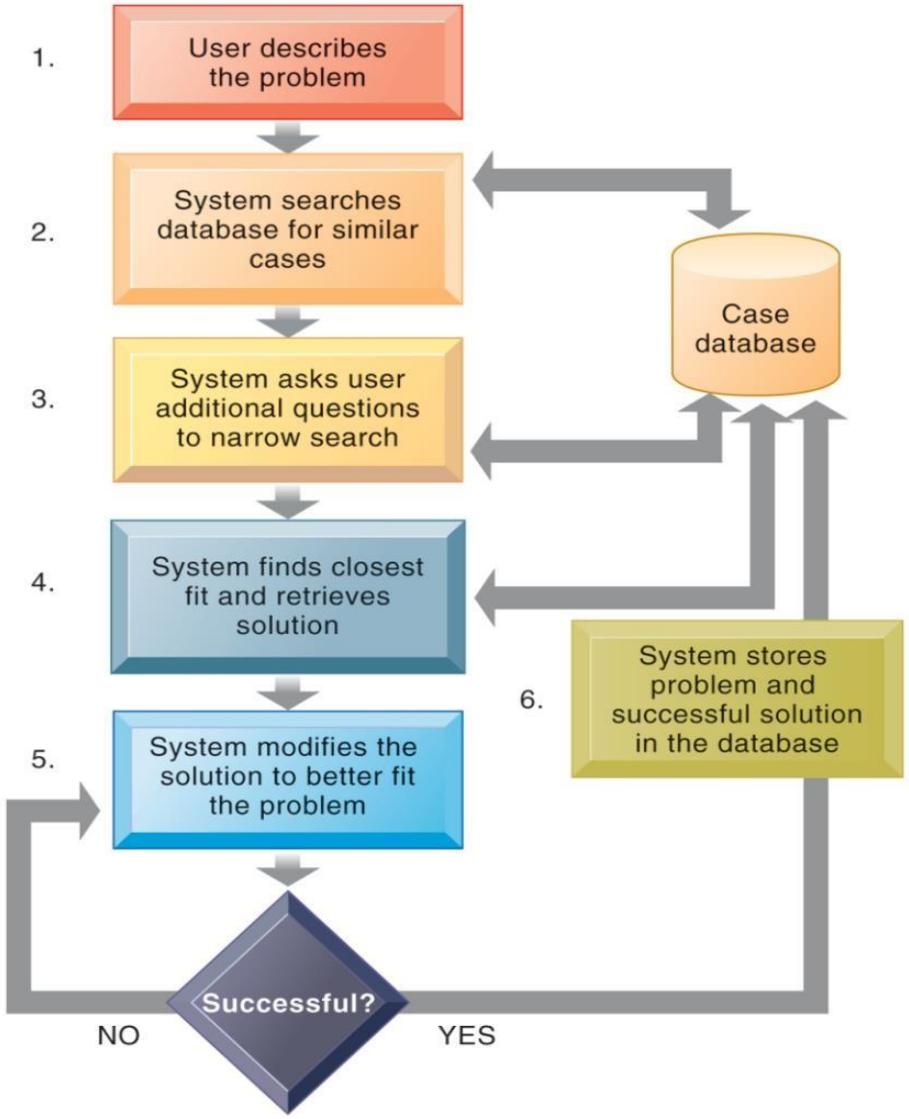
- Most expert systems deal with problems of classification
  - Have relatively few alternative outcomes
  - Possible outcomes are known in advance

- Many expert systems require large, lengthy, and expensive development and maintenance efforts
  - Hiring or training more experts may be less expensive

Case-based Reasoning (CBR)

- Descriptions of past experiences of human specialists (cases), stored in knowledge base
- System searches for cases with characteristics similar to new one and applies solutions of old case to new case
- Successful and unsuccessful applications are grouped with case
- Stores organizational intelligence: Knowledge base is continuously expanded and refined by users
- CBR found in
  - **Medical diagnostic systems**
  - **Customer support**

How Case-based Reasoning Works

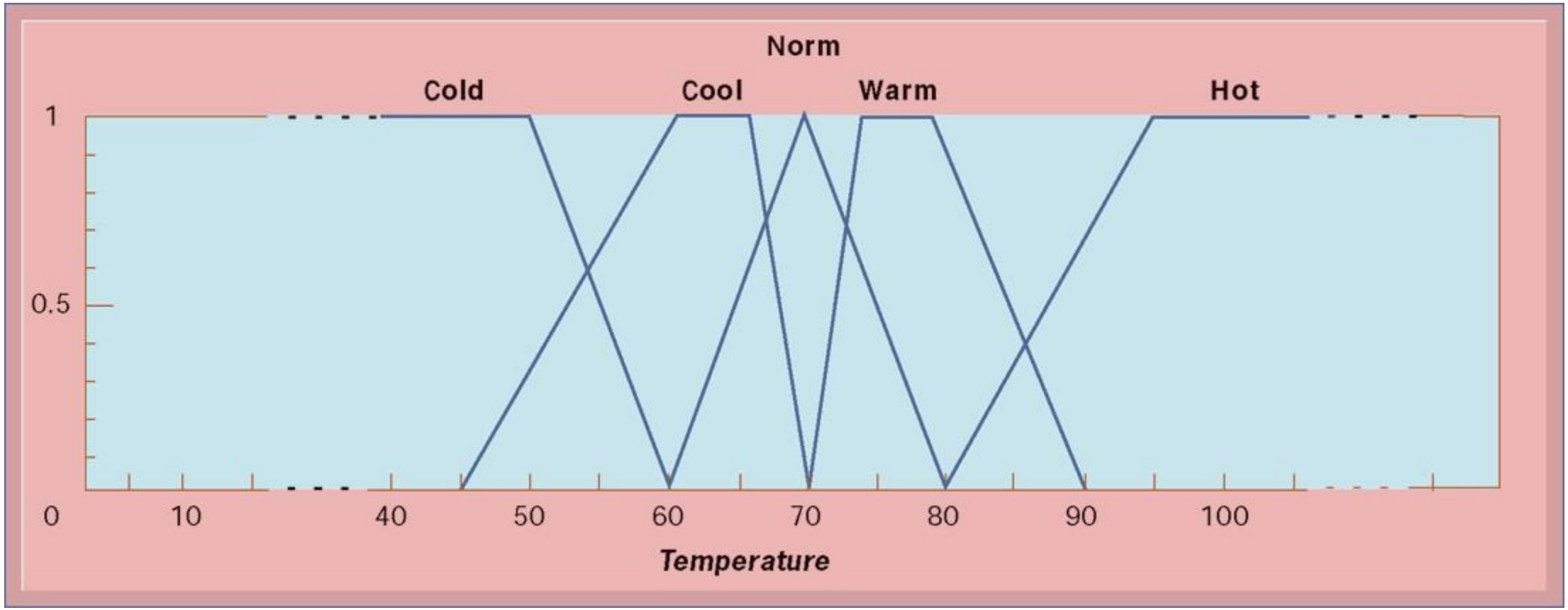


Case-based reasoning represent knowledge as a database of past cases and their solutions. The system uses a six- step process to generate solutions to new problems encountered by the user.

Fuzzy Logic Systems

- Rule-based technology that represents imprecision used in linguistic categories (e.g., “cold,” “cool”) that represent range of values
- Describe a particular phenomenon or process linguistically and then represent that description in a small number of flexible rules
- Provides solutions to problems requiring expertise that is difficult to represent with IF-THEN rules
  - Autofocus in cameras
  - Detecting possible medical fraud

Fuzzy Logic for Temperature Control



The membership functions for the input called temperature are in the logic of the thermostat to control the room temperature. Membership functions help translate linguistic expressions such as warm into numbers that the computer can manipulate.

Machine Learning

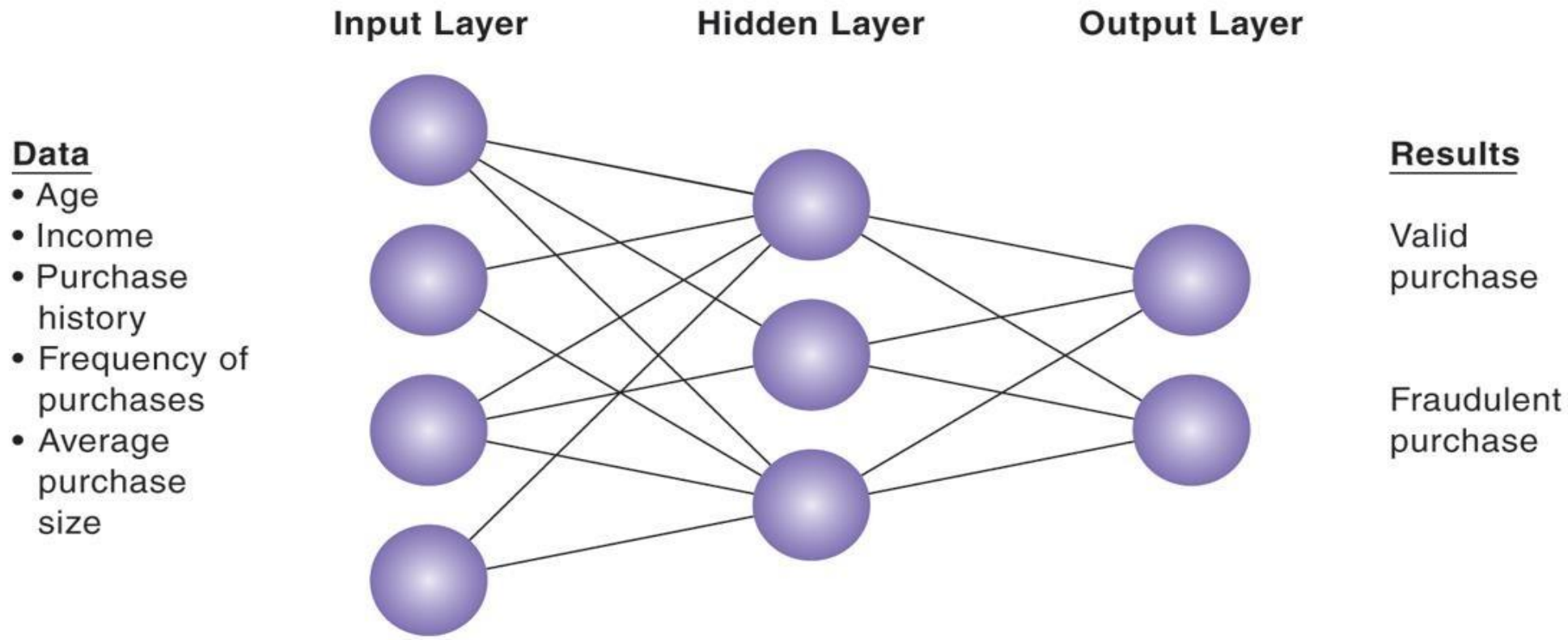
- How computer programs improve performance without explicit programming
  - Recognizing patterns
  - Experience
  - Prior learnings (database)

- Contemporary examples
  - Google searches
  - Recommender systems on Amazon, Netflix

Neural Networks

- Find patterns and relationships in massive amounts of data too complicated for humans to analyze
- “Learn” patterns by searching for relationships, building models, and correcting over and over again
- Humans “train” network by feeding it data inputs for which outputs are known, to help neural network learn solution by example
- Used in medicine, science, and business for problems in pattern classification, prediction, financial analysis, and control and optimization

How a Neural Network Works



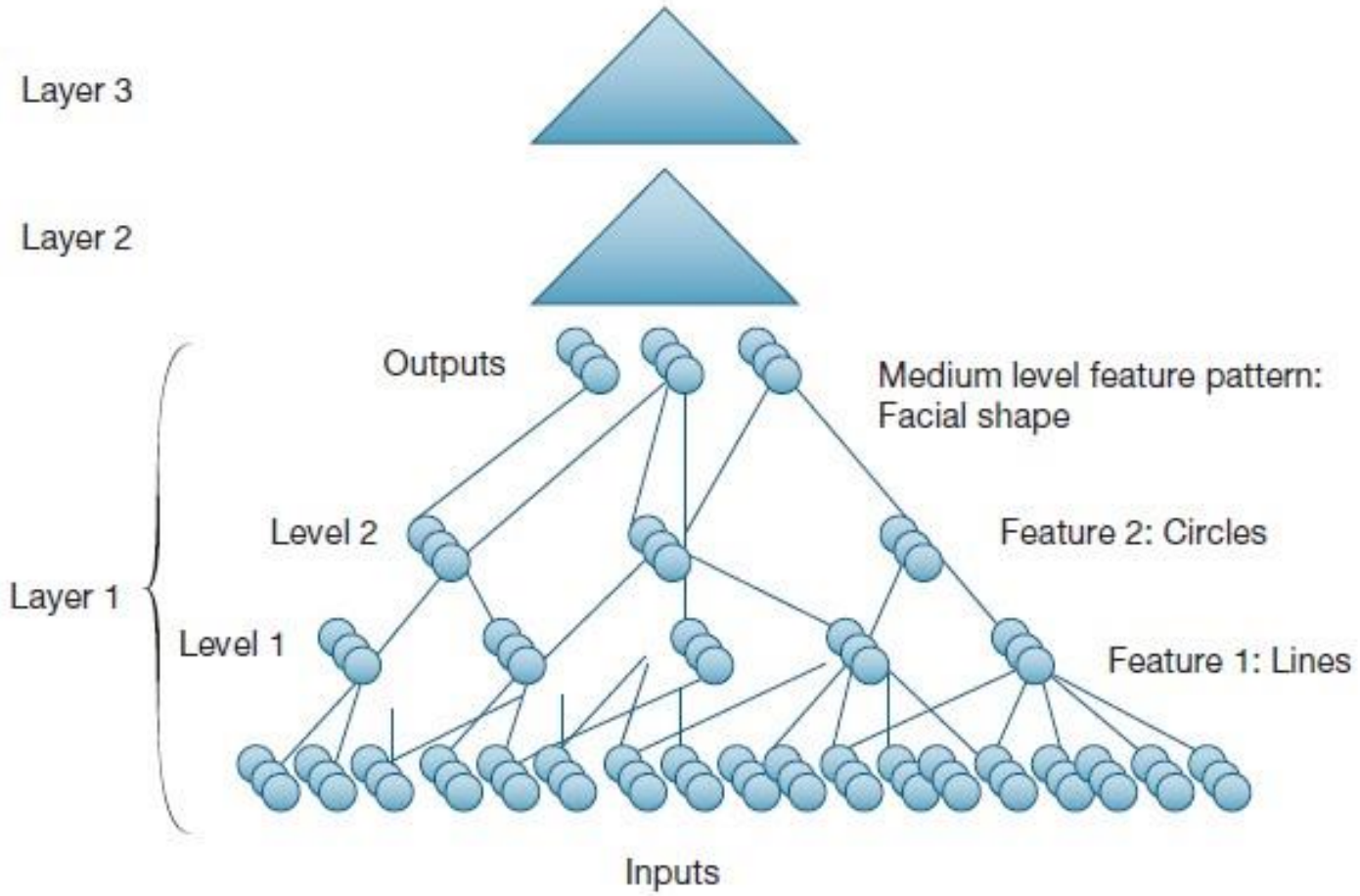
A neural network uses rules it "learns" from patterns in data to construct a hidden layer of logic. The hidden layer then processes inputs, classifying them based on the experience of the model. In this example, the neural network has been trained to distinguish between valid and fraudulent credit card purchases

“Deep Learning” Neural Networks

- “Deep learning” neural networks are more complex, with many layers of transformation of the input data to produce a target output.
- Collections of neurons are called nodes or layers.
- Deep learning networks are in their infancy and are used almost exclusively for pattern detection on unlabeled data where the system is not told what to look for specifically but to simply discover patterns in the data.

“Deep Learning” Neural Networks

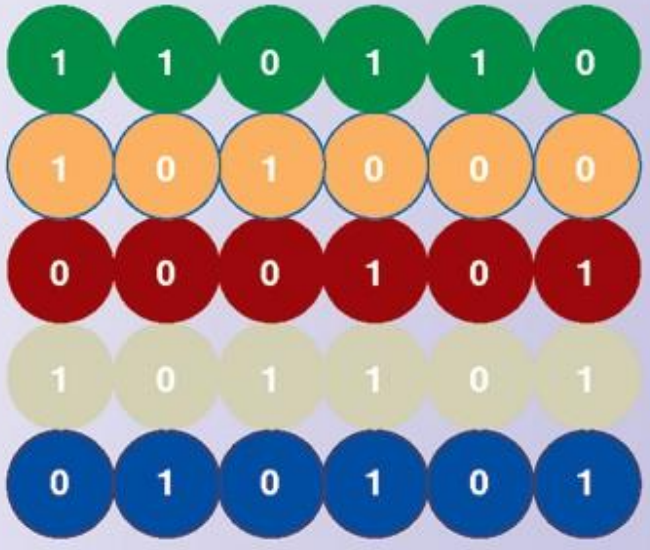
Deep learning networks consist of many layers of neural networks working in a hierarchical fashion to detect patterns. Shown here is an expanded look at layer 1. Other layers have the same structure.



Genetic Algorithms

- Useful for finding optimal solution for specific problem by examining very large number of possible solutions for that problem
- Conceptually based on process of evolution
  - Search among solution variables by changing and reorganizing component parts using processes such as inheritance, mutation, and selection
- Used in optimization problems (minimization of costs, efficient scheduling, optimal jet engine design) in which hundreds or thousands of variables exist
- Able to evaluate many solution alternatives quickly

The Components of a Genetic Algorithm



	Length	Width	Weight	Fitness
1	Long	Wide	Light	55
2	Short	Narrow	Heavy	49
3	Long	Narrow	Heavy	36
4	Short	Medium	Light	61
5	Long	Medium	Very light	74

A population of chromosomes **Decoding of chromosomes** **Evaluation of chromosomes**

This example illustrates an initial population of "chromosomes," each representing a different solution. The genetic algorithm uses an iterative process to refine the initial solutions so that the better ones, those with the higher fitness, are more likely to emerge as the best solution.

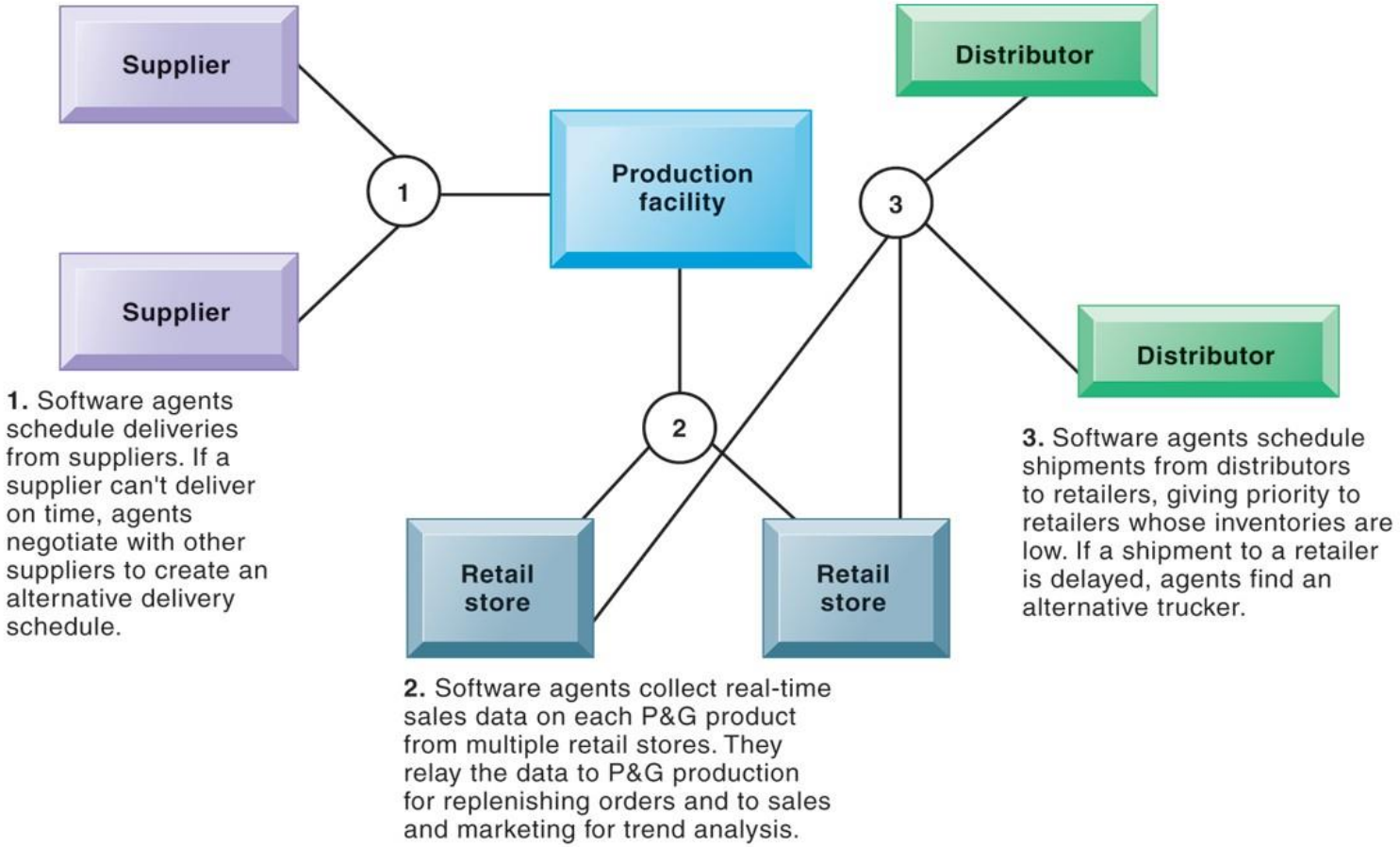
Natural Language Processing, Computer Vision Systems, and Robotics

- Natural language processing (NLP) makes it possible for a computer to understand and analyze natural language.
- Computer vision systems deal with how computers can emulate the human visual system to view and extract information from real-world images.
- Robotics deals with the design, construction, operation, and use of movable machines that can substitute for humans along with computer systems for their control, sensory feedback, and information processing.

Intelligent Agents

- Work without direct human intervention to carry out specific, repetitive, and predictable tasks for user, process, or application
  - Deleting junk e-mail
  - Finding cheapest airfare
- Use limited built-in or learned knowledge base
  - Some are capable of self-adjustment, for example: Siri
- Agent-based modeling applications:
  - Systems of autonomous agents
  - Model behavior of consumers, stock markets, and supply chains; used to predict spread of epidemics

Intelligent Agents in P&G's Supply Chain Network



Intelligent agents are helping P&G shorten the replenishment cycles for products such as a box of Tide.

Hybrid AI Systems

- Genetic algorithms, fuzzy logic, neural networks, and expert systems integrated into single application to take advantage of best features of each

- For example:
  - Matsushita “neurofuzzy” washing machine that combines fuzzy logic with neural networks

Questions ???



Thank you…